

FUNCTIONS

BEGINNER LEVEL

1. Write a program to calculate the sum of all elements in a list.
2. Write a program to reverse a list using slicing.
3. Write a program to convert a list into a tuple.
4. Write a program to access the 3rd element of a tuple.
5. Write a program to convert a list into a set and remove duplicates.
6. Write a program to add an element to a set and remove another element.
7. Write a program to create a dictionary with keys: name, age, and course.
8. Write a program to update the value of a key in a dictionary.
9. Write a program to check if a given number exists in a list.
10. Write a program to check if a given key exists in a dictionary.
11. Write a program to reverse a string without using built-in functions.
12. Write a program to count the number of vowels in a string.
13. Write a program to find the longest word in a sentence.

INTERMEDIATE LEVEL

14. Write a program to find the shortest word in a sentence.
15. Write a program to count words in a sentence having more than 5 letters.
16. Write a program to reverse each word in a sentence while keeping word order.
17. Write a program to count frequency of each character in a string.
18. Write a program to remove all vowels from a string.
19. Write a program to find intersection of two lists without using sets.
20. Write a program to remove duplicates from a list while preserving order.

ADVANCED LEVEL

21. Write a program to find the second largest number in a list.
22. Write a program to merge two dictionaries and sum values of duplicate keys.
23. Write a program to check if two strings are anagrams.
24. Write a program to find all unique pairs in a list that sum to a target value.
25. Write a program to check if a string has balanced parentheses.
26. Write a program to flatten a nested list.
27. Write a program to transpose a matrix (list of lists).